

INSURANCE PORTFOLIO ANALYTICS

End-to-End SQL & Power BI Project Report

Portfolio Performance • Claims Risk • Management Actions

Reporting period: 2022–2024

Project documentation

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1. Executive Summary

This project develops an end-to-end insurance portfolio analytics solution using SQL and Power BI. It evaluates premium performance, incurred claims, loss ratios, claim frequency, claim severity and portfolio exposure, then translates the findings into practical management actions.

Across 2022–2024, the portfolio generated approximately £105 million in premium and £74 million in incurred claims, with an overall loss ratio of 70.29% across approximately 354,000 policy records. The analysis identified a material deterioration in 2024 and showed that performance varied significantly by policy type and other risk segments.

Portfolio headline metrics

Total Premium	Incurred Claims	Overall Loss Ratio	Policy Records
£105M	£74M	70.29%	354K

The report is organised into three complementary Power BI pages: an Executive Overview, a Risk & Claims Analysis page, and a Management Action Matrix. Together they move from portfolio-level performance to claims-driver diagnosis and decision support.

2. Business Problem & Project Objective

Business Problem

An insurance portfolio can generate significant premium income while still experiencing deteriorating profitability if claims costs, loss ratios and risk concentrations are not monitored effectively. Management therefore needs a clear view of portfolio performance across policy types, vehicle-age segments and business categories to identify areas contributing disproportionately to claims and underwriting risk.

Project Objective

The objective is to develop an end-to-end insurance portfolio analytics solution using SQL and Power BI to evaluate premium performance, incurred claims, loss ratios, claim frequency, claim severity and portfolio exposure. The analysis identifies high-risk segments, evaluates changes in portfolio performance over time, and translates analytical findings into management actions such as Review, Monitor, Maintain and Grow to support data-driven underwriting and portfolio management decisions.

3. Technical Architecture & Workflow

Data Source → SQL Data Preparation → Analytical Views → Power BI Data Model → DAX Measures → Dashboards → Management Actions

The project followed an end-to-end analytics workflow designed to separate data preparation, analytical calculation and business reporting. Source insurance data was prepared and analysed in SQL, where data-quality checks, transformations and analytical views were developed to support portfolio, claims and risk analysis.

The analytical views were imported into Power BI, where additional DAX measures were developed for key insurance metrics including Loss Ratio, Claim Frequency, Claim Severity and Exposure Share.

Executive Overview: Overall 2022–2024 portfolio performance and 2024 risk segmentation.

Risk & Claims Analysis: Detailed analysis of claim frequency, claim severity and policy-type risk.

Management Action Matrix: 2024 policy-type exposure, profitability and risk metrics translated into management classifications.

4. Dataset & Data Preparation

The project uses motor insurance portfolio data covering the 2022–2024 reporting period. The dataset contains information required to analyse portfolio exposure, premium income, claims experience and underwriting performance across several risk dimensions.

Key analytical dimensions include policy type, vehicle age, business type and reporting year. Principal quantitative fields include premium, incurred claims, claim counts, exposure and policy records.

Data preparation and transformation were performed in SQL before the reporting layer was developed in Power BI. The SQL stage included data-quality checks, aggregation and the creation of analytical views designed for specific portfolio and claims analyses.

Analysis-ready outputs supported

- Portfolio-level yearly performance
- Policy-type performance and exposure analysis
- Claims frequency and severity analysis
- Vehicle-age risk segmentation
- Business-type performance analysis
- Management-level portfolio classification

5. Insurance KPIs & Analytical Measures

Measure	Formula	Business Interpretation
Loss Ratio	$\text{Incurred Claims} \div \text{Premium}$	Measures the proportion of premium consumed by incurred claims. A high or rising ratio may indicate deteriorating underwriting performance.
Claim Frequency	$\text{Number of Claims} \div \text{Exposure}$	Measures how frequently claims occur relative to exposure and helps identify segments generating claims unusually often.
Claim Severity	$\text{Incurred Claims} \div \text{Number of Claims}$	Measures the average cost of claims and helps distinguish high-cost claims from high-frequency claims.
Exposure Share	$\text{Segment Exposure} \div \text{Total Portfolio Exposure}$	Measures the relative materiality of a segment so management actions consider portfolio size as well as profitability.

Frequency and severity are analysed separately because adverse claims performance can arise from more frequent claims, more expensive claims, or a combination of both.

6. Dashboard Analysis & Key Findings

6.1 Executive Portfolio Performance

Across 2022–2024, the portfolio generated approximately £105 million in premium and £74 million in incurred claims, producing an overall loss ratio of 70.29% across approximately 354,000 policy records.

The yearly trend indicates deterioration in claims performance. The portfolio loss ratio increased from approximately 65.62% in 2022 to 66.46% in 2023, before rising more substantially to 74.66% in 2024.

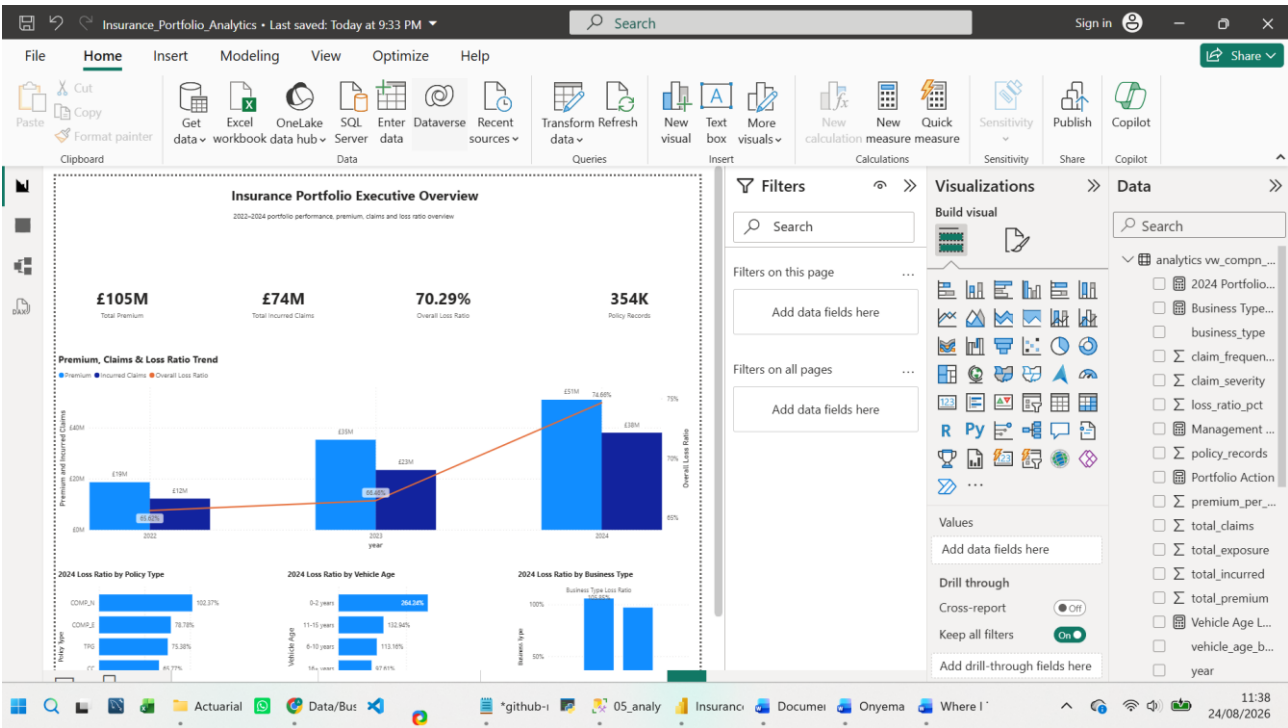


Figure 1. Executive Overview dashboard

6.2 2024 Risk Segmentation

Policy-type analysis identified COMP_N as the weakest-performing segment, with a 102.37% loss ratio, meaning incurred claims exceeded premium. COMP_E and TPG also recorded elevated loss ratios of approximately 78.78% and 75.38%, respectively.

Vehicle-age analysis revealed substantial variation in loss ratios across age bands, indicating that vehicle age is an important risk differentiator. Business-type analysis also showed differences in claims performance between portfolio segments.

6.3 Claims Drivers

Separating claims performance into frequency and severity provided further explanation for policy-type results. COMP_N displayed particularly high claim frequency, indicating that its poor performance is primarily associated with claims occurring more frequently rather than simply having the highest average claim cost.

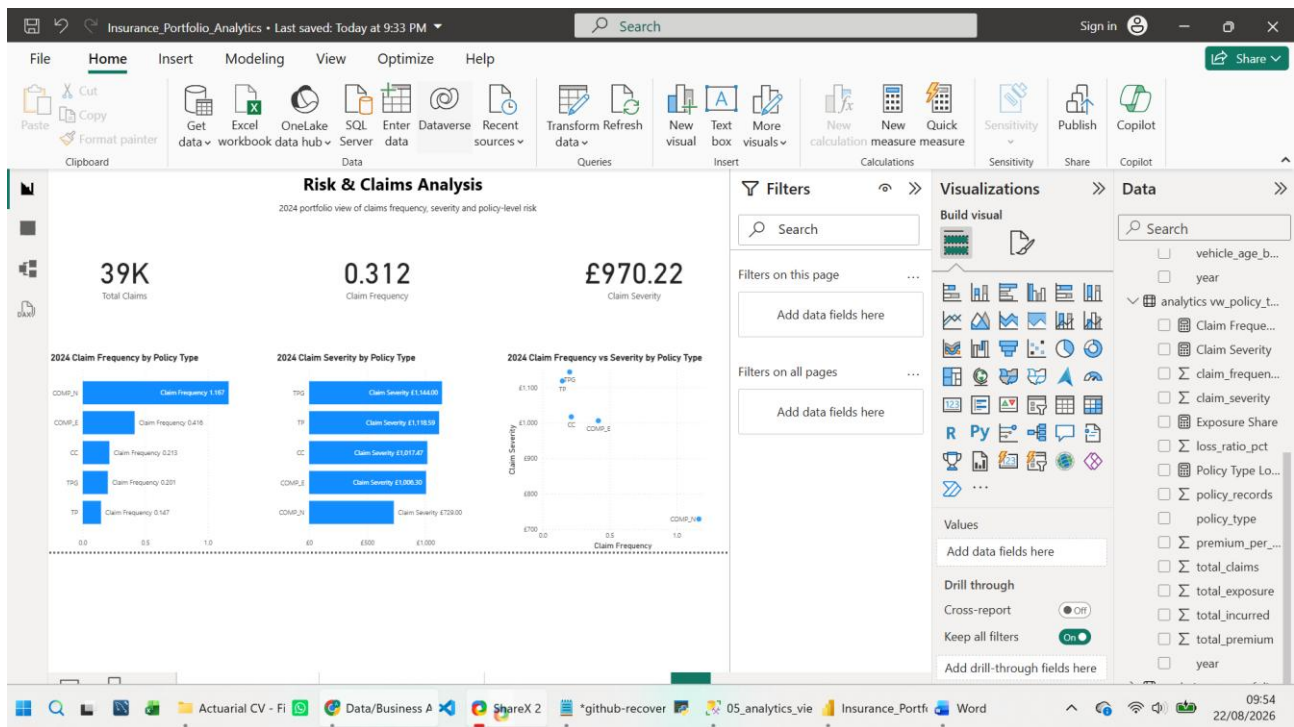


Figure 2. Risk & Claims Analysis dashboard

6.4 Portfolio Concentration

The 2024 analysis identified CC as the largest policy-type segment, representing approximately 56.93% of exposure, followed by COMP_E at approximately 27.16%. This concentration means that even moderate deterioration in CC could materially affect overall portfolio performance. COMP_N represented only approximately 4.79% of exposure, but its adverse loss ratio and high claim frequency make it important for targeted investigation.

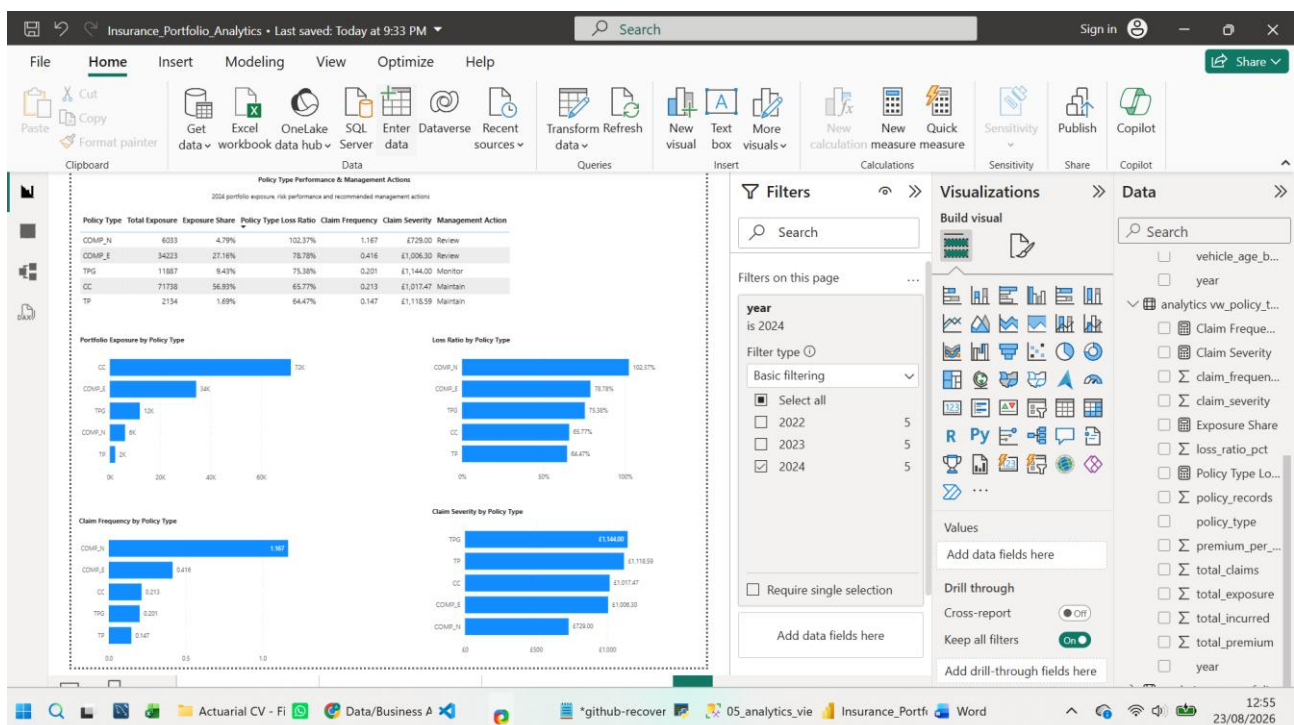


Figure 3. Management Action Matrix dashboard

7. Management Recommendations

Policy Type	Action	Loss Ratio	Exposure Share	Recommended Response
COMP_N	Review	102.37%	4.79%	Review pricing adequacy, underwriting criteria and claims patterns. Investigate whether customer, vehicle, geographic or claims characteristics are driving the elevated frequency before significant growth is pursued.
COMP_E	Review	78.78%	27.16%	Review pricing and underwriting performance. Its material exposure and weaker profitability justify targeted corrective action while preserving valuable business where appropriate.
TPG	Monitor	75.38%	9.43%	Monitor loss ratio, frequency and severity trends closely. Its smaller exposure share supports observation before stronger intervention.
CC	Maintain	65.77%	56.93%	Maintain the segment while monitoring performance carefully because its dominant exposure means deterioration could materially affect the overall portfolio.
TP	Maintain	64.47%	1.69%	Maintain current positioning. The favourable loss ratio is encouraging, but the small exposure base is insufficient evidence for an aggressive growth recommendation.

Portfolio-level recommendation: the 2024 deterioration should not be addressed through a uniform pricing or underwriting response. The analysis supports a segment-specific strategy combining targeted investigation, close monitoring and preservation of better-performing business.

8. Technical Validation, Limitations & Assumptions

8.1 Technical Validation

Quality-control checks were performed across the SQL analytical layer and Power BI report to ensure that key metrics were internally consistent and interpreted within the correct reporting period.

The Management Action analysis was standardised to 2024 so that loss ratio, claim frequency, claim severity, exposure and management classifications were evaluated on a consistent basis.

Exposure calculations were reviewed during validation. Directly summing the raw total_exposure field produced an overstated result because of the underlying granularity of the analytical view. A controlled calculation aggregated exposure once for each policy type, reconciling 2024 policy-type exposure to 126,015 and preventing double counting.

Cross-page comparisons were also performed for key claims measures so that differences caused by filter context were identified and corrected.

8.2 Limitations & Assumptions

- The analysis is based on the variables available in the project dataset and is not a complete insurance pricing or reserving model.
- Review, Monitor, Maintain and Grow are analytical decision-support rules developed for this portfolio project, not a formal insurer underwriting policy.
- A complete profitability assessment may also require expenses, commission, reinsurance, acquisition costs, investment income and capital requirements.
- Frequency and severity results should be interpreted alongside exposure credibility; smaller segments may produce less stable results.
- The analysis identifies associations and portfolio patterns rather than proving causation. More granular policy and claims investigation would be required before material pricing or underwriting changes.

9. Conclusion & Business Impact

This project demonstrates how insurance portfolio data can be transformed into practical management intelligence through an integrated SQL and Power BI analytics workflow.

The analysis identified a clear deterioration in portfolio performance during 2024 and demonstrated that the deterioration was not evenly distributed across the portfolio. Combining loss ratio, claim frequency, claim severity and exposure analysis enabled the project to move beyond headline results and identify the segments contributing most significantly to underwriting risk.

The Management Action Matrix translated these findings into a structured Review, Monitor, Maintain and Grow framework, demonstrating how analytical outputs can support differentiated underwriting, pricing, claims and portfolio-management decisions.

From a technical perspective, the project demonstrates an end-to-end analytical process covering data preparation, SQL analytical views, insurance KPI development, DAX calculations, Power BI visualisation, data validation and business interpretation.

Overall, the project illustrates how insurance analytics can help management move from simply reporting historical results to identifying where performance is deteriorating, what is driving the deterioration, and where further investigation or intervention should be prioritised.

Appendix A — Management Action Logic

Rule	Action
Loss Ratio $\geq$ 100%	Review
Loss Ratio $\geq$ 75% and Exposure Share $\geq$ 10%	Review
Loss Ratio $\geq$ 75%	Monitor
Loss Ratio $<$ 65% and Exposure Share $\geq$ 10%	Grow
Loss Ratio $<$ 65%	Maintain
Otherwise	Maintain